

Project 1 Setup - Group 19

Group Members

1. Krutyanjay Shinde
 - ASU ID :- 1234108189
2. Vaishnavi Sardeshpande
 - ASU ID :- 1233932520
3. Veena
 - ASU ID :- 1232463962

Krutyanjay - Output and summary

```
gem5 Simulator System. http://gem5.org
gem5 is copyrighted software; use the --copyright option for details.

gem5 version 21.0.0.0
gem5 compiled Aug 29 2025 04:11:45
gem5 started Aug 29 2025 12:30:55
gem5 executing on krutyanjay-VirtualBox, pid 10435
command line: build/ARM/gem5.opt ./configs/example/se.py -c tests/test-progs/hello/bin/arm/linux/hello

warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
Global frequency set at 1000000000000 ticks per second
warn: No dot file generated. Please install pydot to generate the dot file and pdf.
warn: DRAM device capacity (8192 Mbytes) does not match the address range assigned (512 Mbytes)
0: system.remote_gdb: listening for remote gdb on port 7000
**** REAL SIMULATION ****
info: Entering event queue @ 0. Starting simulation...
Hello world!
Exiting @ tick 2916500 because exiting with last active thread context
root@krutyanjay-VirtualBox:/home/krutyanjay/Desktop/CSE_520_project/gem5-21.0.0.0#
```

- The gem5 simulator was installed and run within a virtualized environment. The host system is a Windows machine, running a hypervisor called **Oracle VirtualBox**.
- A guest virtual machine was provisioned with the following specifications:
 - **Operating System:** Ubuntu 20.04 LTS
 - **Storage:** 40 GB of virtual disk space
 - **Memory:** 8 GB of RAM
 - **CPU:** 4 cores
 - **Kernel Version:** 6.10.14-linuxkit
 - **gcc Version:** 9.4.0
 - **g++ Version:** 9.4.0
 - **scons Version:** 3.1.2
 - **Python Version:** 3.8.10

Vaishnavi: Gem5 setup using Docker and Summary.

```

gem5-work — root@94682f574a3d: /gem5 — docker run -it -v ~/gem5-work:/workspace -w /gem5 vijayk2003/gem5-custom-...
root@94682f574a3d:/gem5# build/ARM/gem5.opt ./configs/example/se.py -c tests/test-progs/hello/bin/arm/linux/hello
gem5 Simulator System.  http://gem5.org
gem5 is copyrighted software; use the --copyright option for details.

gem5 version 21.0.0.0
gem5 compiled Aug 31 2025 01:19:39
gem5 started Aug 31 2025 01:47:15
gem5 executing on 94682f574a3d, pid 10732
command line: build/ARM/gem5.opt ./configs/example/se.py -c tests/test-progs/hello/bin/arm/linux/hello

warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
Global frequency set at 1000000000000 ticks per second
warn: DRAM device capacity (8192 Mbytes) does not match the address range assigned (512 Mbytes)
0: system.remote_gdb: listening for remote gdb on port 7000
**** REAL SIMULATION ****
info: Entering event queue @ 0. Starting simulation...
Hello world!
Exiting @ tick 2916500 because exiting with last active thread context
Simulated exit code not 0! Exit code is 13
root@94682f574a3d:/gem5# Vaishnavi Sardeshpande (1233932520)

```

The gem5 simulator was installed and run within a Docker container on macOS.

The host system is a MacBook M4 with Apple Silicon. A Docker container was setup with the following specifications:

- **Operating System:** Ubuntu 20.04.5 LTS
- **CPU:** 10 cores (x86_64 architecture is emulated)
- **Memory:** 8 GB RAM
- **Storage:** 453 GB virtual disk
- **Kernel Version:** 6.10.14-linuxkit
- **gcc Version:** 9.4.0
- **g++ Version:** 9.4.0
- **scons Version:** 3.1.2
- **Python Version:** 3.8.10

Upon running the highlighted command line, “Hello world!” is printed which indicates the successful setup of gem5.

Veena- Output and summary

```

veenamadhuri — root@074b2df3faaa: /gem5 — docker run -it vijayk2003/...
gem5 is copyrighted software; use the --copyright option for details.

gem5 version 21.0.0.0
gem5 compiled Sep  3 2025 04:50:12
gem5 started Sep  3 2025 05:13:17
gem5 executing on 074b2df3faaa, pid 11117
command line: build/ARM/gem5.opt ./configs/example/se.py -c tests/test-progs/hello/bin/arm/linux/hello

warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
warn: membus.slave is deprecated. `slave` is now called `cpu_side_ports`
Global frequency set at 1000000000000 ticks per second
warn: DRAM device capacity (8192 Mbytes) does not match the address range assigned (512 Mbytes)
0: system.remote_gdb: listening for remote gdb on port 7000
**** REAL SIMULATION ****
info: Entering event queue @ 0. Starting simulation...
Hello world!
Exiting @ tick 2916500 because exiting with last active thread context
Simulated exit code not 0! Exit code is 13
root@074b2df3faaa:/gem5#

```

The gem5 simulator was successfully installed and executed within a Docker container on macOS. The host system is a MacBook M4 powered by Apple Silicon, and the Docker container was configured with the following specifications:

- OPERATING SYSTEM/SYSTEM VERSION : Ubuntu 20.04.5 LTS
- CPU CORES : 8 CORES
- MEMORY : 8 GB of RAM
- STORAGE : 324 GB
- Kernel Version : 6.10.14-linuxkit
- Python Version : Python 3.8.10
- GCC/g++ Version : gcc (Ubuntu 9.4.0-1ubuntu1~20.04.1) 9.4.0